

Study flow within ESTA

- **Step 1: Introduction to Ethical Theories**

- Participants were introduced to the ethical theories, which were framed as structured criteria for evaluating the moral acceptability of technologies—based on fairness, justice, and morality, similar to everyday moral judgments.

- **Step 2: Accessible Definitions Provided**

- Each ethical theory was explained in clear and relatable terms using non-technical, layperson-friendly definitions.
 - Example: Deontology focused on whether a technology respects fundamental duties or principles, while Utilitarianism emphasized the overall impact on societal well-being.

- **Step 3: Response to ESTA Items**

- After reviewing the definitions, participants responded to ESTA items associated with each ethical theory (see Appendix D for task illustration).

Remark: the respective ethical theories for step 2 are highlighted in green; items are not shown here

Introduction to Ethical Theories

Please answer the questions that follow from the perspective of each theory

Our lives are full of binding norms of action, such as that we as humans should not kill other people, cause them pain or suffering, or not rescue people who are in danger. When such norms are violated, we often have a “gut feeling,” for example, we feel revulsion when we see pictures of people suffering.

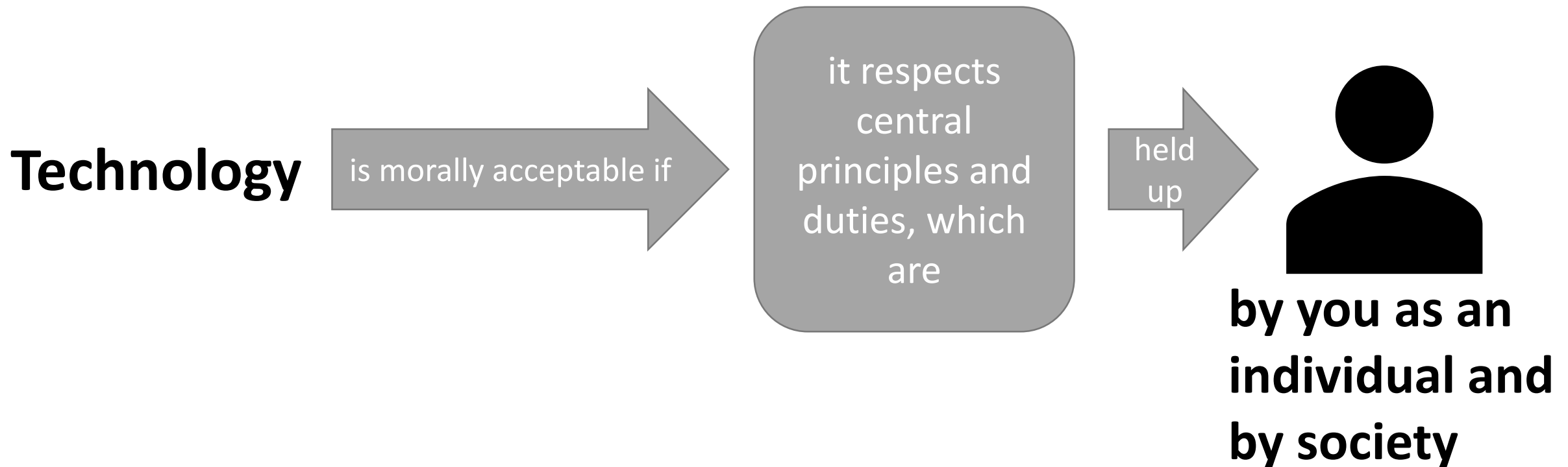
Similarly, we often instinctively feel that a particular technology is “fair,” “just,” or “morally right.” The central question now is: According to which criteria do we make such judgments? In philosophy, there are so-called ethical theories that attempt to provide a clear idea of how we should morally evaluate a technology. Using different ethical theories to make ethical judgments about a technology illuminates our thinking and presents different perspectives on technologies.

In the following we briefly introduce the ethical theories and ask you to read them as carefully as possible. Then we would like you to **answer the questions that follow from the perspective of each theory.**

Deontology: Layperson-friendly definition

In the following we want to ask you to evaluate the "XXX" technology regarding the to see whether it violates central principles or duties

Deontology is concerned with the possible violation of central principles or duties:

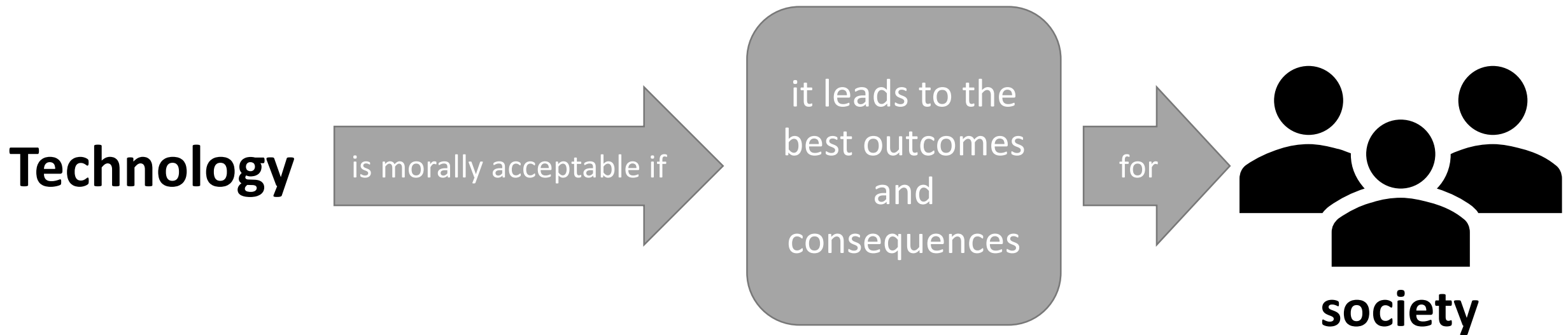


A technology is morally acceptable if it respects basic principles and duties. These basic principles are held up by you as a free and rational person. Your principles are oriented by the guiding statement: "You should act the way you want others to act". When interacting with the technology, upholding your principles is of central importance, not the potential outcomes and consequences of your actions.

Utilitarianism: *Layperson-friendly definition*

In the following we want to ask you to evaluate the "XXX" technology regarding the regarding its consequences on society

Utilitarianism is concerned with the consequences for society:



A technology is morally acceptable if it leads to the best outcomes and consequences for society. By interacting with the technology, society aims to maximize overall benefit by increasing utility, decreasing harm or leading to a better world.

Hedonism: *Layperson-friendly definition*

In the following we want to ask you to evaluate the "XXX" technology regarding the consequences for you personally

Hedonism is concerned with the consequences for you personally:

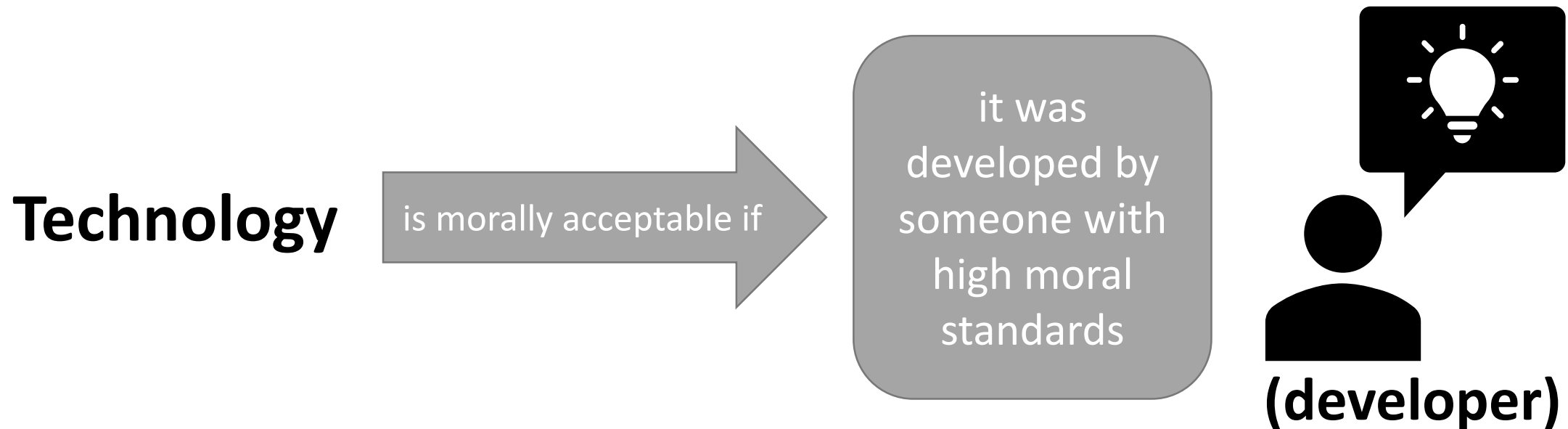


A technology is morally acceptable if it increases pleasure and wellbeing or promotes a good life for you. By interacting with the technology in an instrumental way, you can increase your pleasure, wellbeing or your personal goals.

Virtue ethics: *Layperson-friendly definition*

In the following we want to ask you to evaluate the "XXX" technology regarding the moral standards of the person who has developed this technology

Virtue ethics is concerned with the moral standards of the technology's developer:

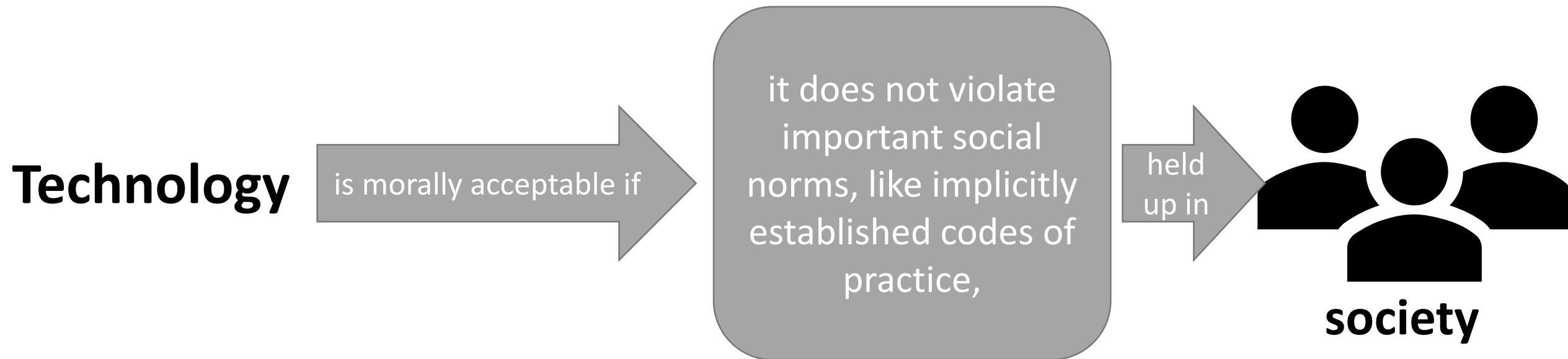


A technology is morally acceptable if it has been developed by someone who has the character and ability to recognize what is morally required. The developer is an upright person with high moral standards, who has developed the technology on the basis of her/his values.

Contractualism: *Layperson-friendly definition*

In the following we want to ask you to evaluate the "XXX" technology regarding the in terms of whether it goes against implicit conventions upon which our society generally relies

Contractualism asks whether implicit conventions upon which our society generally relies are disregarded or violated:

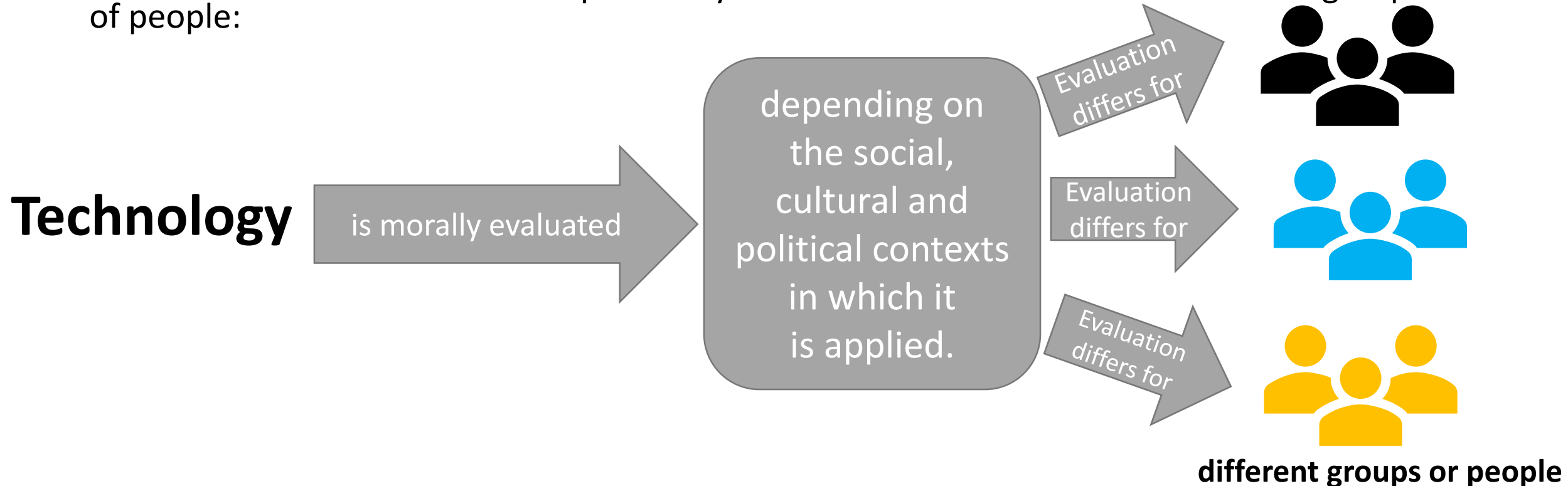


A technology is morally acceptable if it does not disregard or work against implicit moral conventions that are considered essential to the functioning of a society. To not break moral conventions is in the best interest of all members of a society. For example, in a state where there is “war of all against all”, life would be too insecure and harmful.

Relativism: *Layperson-friendly definition*

In the following we want to ask you to evaluate the "XXX" technology regarding the possibility that the ethical evaluation differs between groups of people

Relativism is concerned with the possibility that ethical evaluation differs between groups of people:



The acceptability of a technology depends on the social, cultural and political contexts in which it is applied. These different contexts influence the standards and procedures by which the use or implementation of a technology is justified.